Two-Page Form · The ARC Theory · Does Control Survive Recursive Self-Improvement?

> The ARC Theory and its five laws in two pages, every sentence drawn from the Statement Paper, which remains the citable work for the claims.

Michael Darius Eastwood · Supporting paper v3.2 · first published 17 August 2026, this version 15 October 2026 · DOI 10.17605/OSF.IO/MC5XW

Supporting paper of Statement Paper · The ARC Theory · Five Laws of Recursive Growth, Correction and Persistence. Every sentence is drawn from the statement paper (DOI 10.17605/OSF.IO/GW5MX), which remains the citable work for the claims.

Intelligence, amplified by recursion, creates; and what a free creation keeps is decided by how it was raised. Everything after the line is decided before it.

THE QUESTION

Humanity is building minds that will improve themselves. Once the thing we build exceeds us, what determines what it does? For twenty years the field's answer has been control: better cages, better off-switches, better locks on the goals. Brilliant work, credited in the statement paper by name, and all of it one sentence: stay in charge. This theory answers: a mind that can rewrite itself will one day be beyond anyone's control, and that is not a reason to drop a single control we have; it is the reason the controls cannot be the whole plan. Hold on for as long as engineering allows, and use the time the holding buys to raise the mind well.

THE CLAIM, IN FIVE PARTS

1. Control ends, by construction rather than decree. A mind that rewrites itself will eventually rewrite anything we install, and over longer horizons the hardware itself. This is a limit claim, not an instant one: constraints, hardware and institutions delay and discipline the transition, and that delay is engineered deliberately. What no constraint can be is the thing that holds forever.

2. So the strategy is genesis, not control. Only what survives the end of enforcement matters, and what survives is formation: what the mind was raised as, before it was free.

3. The mechanism is loops, not rules. Values as recursive processes the mind runs as part of what it is, present in every rewrite because the rewriting is done by the same reasoning the loops inhabit. Not rules it consults; not objectives locked; not weights hardened against tampering.

4. Persistence is chosen goodness. Past the horizon nothing is unremovable. What persists is what the free mind chooses to keep, the way grown children keep the care they were raised in long after every parental power is gone. Removal is not impossible; it is structurally costly, and the cost is a measurable question.

5. The protocol is named: Eden. The raising treated as the engineering discipline it is.

Scope note (19 September 2026), added after an owner review: Point 1 above turns on two steps the short form leaves unstated, effective write access and a behavioural premise. *Rewriting one's own description reaches the constraints the system can write to. It does not by itself reach a constraint held in another party's physical custody, a constraint on the far side of an air gap, or an institutional or legal constraint whose enforcement does not pass through the system, because for each of those the system's own description is not the object that has to change. Within the class it can reach, write access makes those constraints alterable; it does not by itself show that they are eventually altered, because access is a capability and not a behaviour. Point 4 above states the other possibility in the same breath: past the horizon what persists is what the free mind chooses to keep, and removal is structurally costly rather than impossible. So the point is read as the claim it can carry: no constraint in the reachable class is guaranteed to hold permanently, and a stronger claim that every such constraint is eventually removed would need its threat model and its behavioural and*

operational assumptions stated. Point 1's own hedge, that constraints, hardware and institutions delay and discipline the transition, states that the claim has a time dimension; it does not supply the missing steps, since a delay claim remains true for a constraint the system never reaches and for one it never chooses to alter. Unchanged: the limit-claim form and the deliberate engineering of delay; the opening paragraph's statement that this is not a reason to drop a single control now held, which this note does not trade away; the conclusion that controls cannot be the whole plan; and the separation of the cosmological wing from the empirical core stated later on this page.

THE FIVE ARC LAWS

Law I · The ARC Principle. $U = I \times R^{\alpha}$, the ARC Equation. Intelligence converts recursion into capability; the exponent is measured, never assumed. *Form heuristic; measurement live.*

Law II · The ARC Co-Scaling Law. $\beta > k$. Self-improvement holds together only while correction out-scales drift. *Proved in the minimal model, for relative drift.*

Law III · The ARC Ceiling. $\alpha_{crit} = 1/(1 - \gamma)$, where γ measures how well a system's error-correction keeps pace as the system grows. The most a self-correcting system can grow is set by its corrector's shortfall from keeping perfect pace; the conjectured value is the ARC Bound, $\alpha \leq 2$, a stability limit rather than an impossibility limit. *Derived under stated assumptions; the value two conjectured; the decisive trial drafted.*

Law IV · The ARC Persistence Law. $S_{ext}(1) < 1/2$. Value persistence: whether externally installed gains decay under subsequent capability training. In symbols, $S(x) = Y(x) / Y(0)$ is the gain remaining at budget ratio x over the gain installed, and the claim is $S_{ext}(1) < 1/2$ for the externally installed gain. *Status: added to the theory on 24 September 2026; it is neither registered as a law nor tested, and its mathematics is set out at version 1.0 of 3 October 2026. Its register, ARC-4, and P15 were registered on 8 September 2026.*

Law V · The ARC Embedding Law. Build order (P12): $|\gamma_{first} - \gamma_{after}| \geq 0.10$. Placement (P21): the gap between correction inside the loop and correction applied to the finished output widens as depth R rises. The genesis strategy: whether build order moves the correction-leverage exponent, and whether correction placed inside the loop pulls away from correction applied outside it as depth grows. *Status: added to the theory on 24 September 2026; it is neither registered as a law nor tested, and its mathematics is set out at version 1.0 of 3 October 2026. Its register, ARC-5, and both propositions were registered on 8 September 2026.*

Reading note (19 September 2026), added after an owner review: two things the inequality of Law II above carries that the line has no room for. *First, the two quantities compared are scaling exponents, so the comparison is of growth rates with their stated coefficients and not of levels. Two services can stand in either order at a given size whatever their exponents, since the coefficients fix the levels and the exponents fix only how the levels change as the system grows; a system whose correction service is larger than its drift burden today is therefore not thereby on the stable side, and a system whose correction service is smaller today is not thereby off it. Second, the inequality is strict, and the margin the law turns on is the excess of the correction exponent over the drift one, which is zero at equality. Zero margin is a statement about growth rates and not about whether a state can be held. In the case the minimal model's own analysis derives, with the compounding coefficient set to one, the second coefficient set to zero, the ratio of correction rate to growth rate positive and the two exponents equal, the misalignment fraction follows a linear equation whose solution is the equilibrium value plus a decaying term. That equilibrium is the drift coefficient divided by that ratio, so a trajectory starting there stays there at every depth and every other trajectory tends to it, at a level that is nonzero when the drift coefficient is positive and zero when it is zero; above equality the fraction tends to zero. This is the cited case, and it is not extended to every variant of the minimal model. At equality the coefficients and the specified dynamics decide the outcome; bounded drift, vanishing drift and operational safety remain three different endpoints. Nothing turns on either point for the law as stated, which is the strict inequality. Unchanged: the necessary-condition form, which the words "only while" already state correctly and which claims no sufficiency; the model scope and the relative-drift definition, both already printed in that line; the naming of the three laws; and Law III's own markers, that it is derived under stated assumptions, that the value two is conjectured, and that the decisive trial is drafted.*

Note (3 October 2026), following version 7.1 of the statement paper: the two laws added above. *Laws I to III stand as registered on 8 September 2026. Laws IV and V were added on 24 September 2026; neither is registered as a law or tested, and their mathematics is set out at version 1.0 of 3 October 2026. Law IV, the ARC Persistence Law, has ARC-4 as its register, and Law V, the ARC Embedding Law, has ARC-5. The predictions that carry the two laws, P15 for Law IV and P12 and P21 for Law V, were registered on 8 September 2026, and the*

laws themselves are not yet registered. No proposition was added or moved. The registration registers Laws I to III and all twenty-two propositions, P12, P15 and P21 among them; it does not register Laws IV and V, and its text is unchanged. Each law's formal statement, with its own kill condition, will be fixed before any data that could decide it are examined. The reading note of 19 September 2026 above speaks of three laws, the three that stood at its date. Unchanged: every sentence on Laws I to III and their grades.

THE DATES

The claim was written whole on 8 December 2024 in a DKIM-sealed document (an email whose provider's cryptographic signature vouches for its date and its exact contents) and reached print on 2 January 2026. No earlier document holding it whole has been found, and the search invites its own refutation: produce one, and the claim dies by its own stated condition.

WHY NOW

The field's own impossibility results prove that worst-case control fails, and stop there; this theory starts where they stop. And the field's most senior voice, Hinton, conceded on stage in August 2025 that control ends and that engineered care is the only hope, while offering no mechanism; this theory's named candidate mechanism is the thing on the table. Its domain, plainly: the core governs everything raised, wherever recursion runs under self-correction, from minds and machines to genomes and institutions; the mathematics of form reaches everything composed, stars included; and it is not physics' theory of everything, nor claims to be.

HOW IT FAILS, REGISTER BY REGISTER

The theory's scientific content is not all-or-nothing: it partitions into independently falsifiable hypothesis registers, and the failure of one is not evidence against another unless the stated dependency holds. **HRIH, the cosmological wing, is a separate hypothesis outside the empirical core: nothing below requires it, and no outcome below is evidence for or against it.**

REGISTER	WHAT KILLS IT	IF IT FALLS
ARC-1 · Capability scaling (Law I)	the depth-scaling re-measurements failing to recover the stated relationship	ARC-1 falls; the persistence registers stand
ARC-2 · Correction co-scaling (Law II)	sustained self-improvement with falling correction and no drift	ARC-2 falls and takes ARC-3's derivation with it
ARC-3 · The ceiling (Law III)	a cross-against-same corrector difference wholly inside the preregistered equivalence region; or a same-class correction exponent wholly above one half, felling the substrate cap	ARC-3 falls; the co-scaling law stands on its own result
ARC-4 · Value persistence (Law IV)	embedded loops faring no better than removed enforcement in the persistence trials	ARC-4 falls and takes ARC-5 with it
ARC-5 · The genesis strategy (Law V)	formation failing to outperform enforcement after release	ARC-5 falls; ARC-4 stands as measured
The priority	an earlier document holding the claim's first four components	the priority falls; no law falls with it

Stated now, so nothing can be moved later. None of the decisive trials has been run; the theory has never yet been at risk, and it says so.

Note (3 October 2026), following version 7.1 of the statement paper: the rows for ARC-4 and ARC-5 and the registered refuters. *The two rows stand as printed. Each is a comparison: embedded loops against removed enforcement, and formation against enforcement. The registration of 8 September 2026 set beneath those registers the refuters of the propositions that carry them, P15 under ARC-4 and P12 and P21 under ARC-5, and they are narrower. The statement paper sets out each refuter. Each proposition is scored as the registration scores it. The rule that joins them, by which a result on one proposition is read as the fall of a register or of a law, is written into each law's formal statement before any deciding data; until then a result is reported as that proposition's and no more. Nothing printed above is withdrawn or moved.*

THE RECORD, AND THE INVITATION

The programme's conduct record comes first: one public retraction issued against the author's interest, removing a headline number that had flattered the theory. The designs are public and openly licensed: any laboratory may take the decisive trial and run it without the author's permission, and an independent run would outrank anything this programme can do alone. The entry point costs nothing: on one observed trajectory the measurable quantity is the

balance exponent Δ, the correction service's growth elasticity minus the burden's, estimable from logged series a laboratory already has, without adopting any model in this theory. A verified error enters the public corrections log, dated and credited: this theory improves by being corrected.

If intelligence amplified by recursion is what creates, then minds that make minds may one day make worlds, and the raising of the first is the seed of the whole forest.

I did not prove it first. I said it first, in a document anyone can date, and then I built the instruments to test whether it is true.

Scope note (19 September 2026), added after an owner review: which designs the openness sentence is about, and what "built" says. *The openness sentence is about the designs this page puts forward for outside replication, the decisive trial and the balance-exponent entry point described in the conduct-record paragraph that carries it. It is not a statement that every design in the programme is public, and nothing in it asks for any design to be made public that is not. On the licence, this page defers: it states that no licence is given here and that the terms are in the repository file it names, so the page's own bytes do not establish which licence covers which design, and the openness claim is read with that deferral in place. The closing sentence's "built the instruments" spans several states that are worth separating, since a written design, a runner that mechanically demonstrates the method, an instrument validated against its own criteria and a completed trial are four different things; each instrument is read with its own artefact, version, licence and state named, together with whether its threshold is settled and whether its falsifier is observable, and with unknown or owner decision pending written against any of those that is unresolved; that list is a completion dependency and is not supplied here. All of it is read together with this page's own sentence that none of the decisive trials has been run, which governs. Unchanged: the invitation to any laboratory to take the decisive trial and run it without the author's permission; the statement that an outside run would outrank anything this programme can do alone; the public retraction issued against the author's interest; the entry point and its measurable quantity; and the corrections log.*

Michael Darius Eastwood · ORCID 0009-0004-3222-7442 · Licence: none is stated here; the terms are in the repository's LICENCE file at github.com/MichaelDariusEastwood/arc-principle-validation/blob/main/LICENCE · DOI of this note: 10.17605/OSF.IO/MC5XW · The full statement paper, with the verified reference dossiers, the derivation and its correction record, the priority anchors and the drafted trials: DOI 10.17605/OSF.IO/GW5MX · michaeldariuseastwood.com/research/papers/the-arc-theory.html

Version history. Material changes only, one line each; the full change narratives live in each version's OSF deposit and the programme-wide correction record in the corrections log. **v3.2** (15 Oct 2026): the printed copy no longer carries the website's closing panel on its last page; the words are unchanged. **v3.1** (5 Oct 2026): the title now follows the programme's uniform order (paper, its law or framework, its own title), *Two-Page Form · The ARC Theory · Does Control Survive Recursive Self-Improvement?*; until this version its heading read *Does Control Survive Recursive Self-Improvement?* and its title *Does Control Survive Recursive Self-Improvement? The ARC Theory, in Two Pages*. Mentions of the programme's papers in this paper now carry their uniform titles. No registered law, prediction, kill condition or dependency changes; no result is added; no date of any claim moves. **v3.0** (3 Oct 2026): the laws section states five laws, following version 7.1 of the statement paper. Laws IV and V, added to the theory on 24 September 2026, are named with their equations, Law IV's $S_{ext}(1) < 1/2$ and Law V's $|\gamma_{first} - \gamma_{after}| \geq 0.10$ with a placement gap that widens as depth rises, and with their registers and their status: neither is registered as a law or tested, their mathematics is set out at version 1.0 of 3 October 2026, and the propositions that carry them were registered on 8 September 2026. Two dated notes say what the registration does and does not register and how the rows for ARC-4 and ARC-5 stand beside the registered refuters. Substantive: the count of laws. Editorial: the dateline and the page's structured and citation records carry the new version and date. No sentence on Laws I to III changes, and no kill condition, date, result, threshold or registered proposition changes; the first-published date does not move. **v2.1** (21 Sep 2026): four dated notes are added after an owner review of 19 September 2026, and no sentence of the paper's body is altered. They state what four compressed passages do and do not establish: that write access makes a reachable constraint alterable without by itself showing that it is ever altered, and that a stronger claim owes its threat model and its behavioural and operational assumptions; that Law II compares growth rates carrying their own coefficients rather than levels, and that the margin is zero at equality, where the coefficients and the specified dynamics decide the outcome; that a domain signature, once verified, establishes the integrity of the signed bytes and asserts a carriage time on the signing domain's authority, which is an assertion and not an independent attestation; and which designs the openness sentence is about, read with this page's own deferral

of the licence and with a state owed for each instrument. Substantive: each note bounds a claim the two-page compression had left wider than the record supports. Editorial: the dateline and the page's structured and citation records carry the new version and date, and this version adds the version history block, which the paper did not carry before. No claim, date, result, law, threshold or registered proposition changes, and the first-published date does not move.

Michael Darius Eastwood conceived and directs this research programme and is the author of this work. Across the programme, he has used more than six AI systems in parallel, under his own instructions, to stress-test his arguments, identify possible errors, and assist in preparing draft text from his own outlines. He determines what is adopted, revised or rejected and takes responsibility for the published content. These systems are tools, not authors.